



Most websites use CSS along with HTML to stand out aesthetically. With CSS, we can add designs to a website to make it look more appealing.



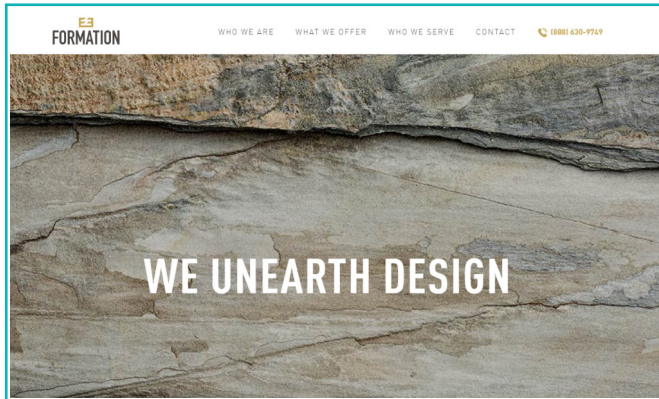
The internet has become an integral part of our lives. We use it to watch videos, read information, talk to people, etc. On a browser, we use websites for these tasks. A website can also act as a platform to display skills, interests, achievements, and thoughts. It can be accessed from anywhere, making the world our audience.

Nowadays, templates are available online to create websites. But readymade templates cannot always suffice requirements. Sometimes we may need to customize a template as per our requirement or even create a website from Scratch without using any pre-built templates. In such scenarios, one must have the knowledge of HTML and CSS.

CSS lets us design it well.

4. Stylesheet

This lesson is an introduction to website creation and design using HTML and CSS. Here are some reference websites:



<http://www.formationstone.com/>



<http://sergiopedercini.com/>

HTML in brief

HTML stands for Hyper Text Markup Language. It is a special (markup) language used to create webpages. A markup language is a computer language that uses tags to define elements within a document. An HTML Webpage can contain text, multimedia files, links, and images. The current version of HTML in use is HTML5.

HTML5



To create a webpage, HTML has pre-defined keywords, called **TAGS**. Every HTML tag has its own purpose and displays content in a specific way.

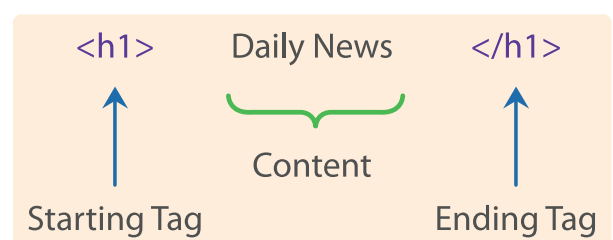
Let us learn a little about HTML.

HTML elements

An HTML element is a component of code in an HTML document which displays content on the webpage. There are different HTML elements for different types of content. It comprises of a start tag and an end tag, with content in between.

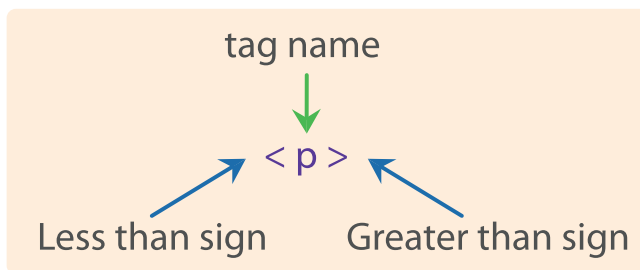
For example:

Heading element h1 is used to display headings.

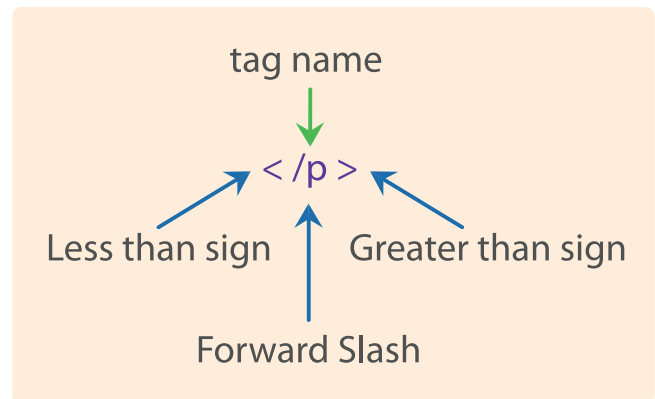


HTML Tags

An HTML tag is a set of characters that make up the start and end of an HTML element. HTML tags have a specific format. This is what their structure looks like:



Opening or start tag



Closing or end tag

We will use some common HTML tags in this lesson:

Tags	Description
<code><html> ... </html></code>	The first tag used at the start of every single webpage.
<code><head> ... </head></code>	The head contains information about a website's properties such as the title, character set, meta-tags and links to external related files
<code><title> ... </title></code>	This tag creates a heading for the page which appears at the top of the browser in the title bar.
<code><body> ... </body></code>	The main tag inside which all HTML code resides.
<code><h1>...</h1>to <h6> ... </h6></code>	This tag is used to create headings with different font sizes.
<code><div> </div></code>	This tag creates an empty container to hold other HTML tags.
<code><p> ... </p></code>	This tag encloses paragraphs on the website.
<code></code>	This tag is used to embed an image on a website.

HTML Closing tags

A closing tag indicates the end of a tag. It is written with a forward slash in the beginning.



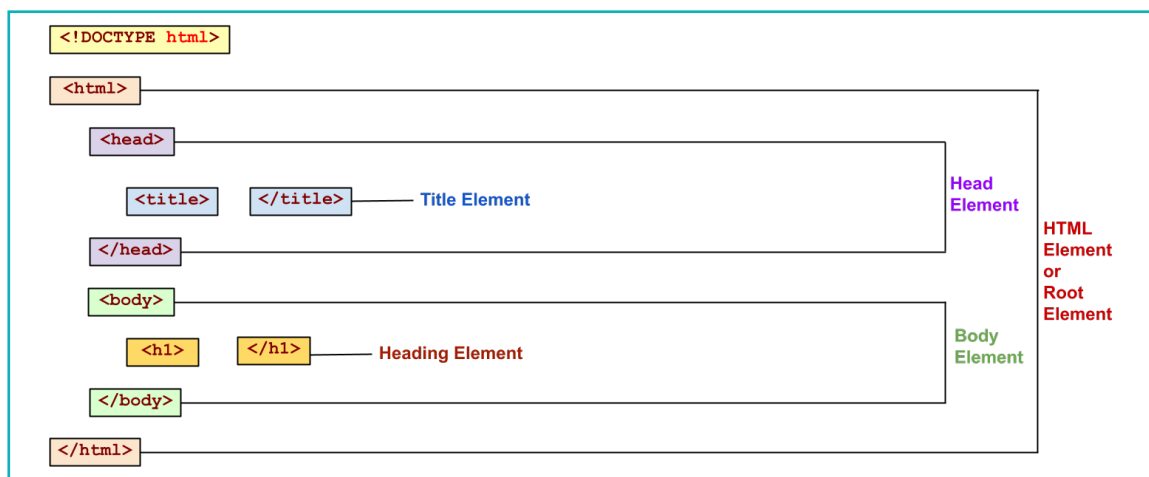
Most elements have an opening and closing tag. There are few exceptions such as `<img>` tag.

Apart from these, HTML has some more tags. For an understanding of all HTML tags, their attributes, properties, and usage, visit the following link:

<https://developer.mozilla.org/en-US/docs/Web/HTML>

Basic structure of an HTML page

Every HTML webpage has a standard webpage structure as explained below:



Software for HTML code

To write HTML and CSS code, we will use a software called **VS Code**, which is a **Code Editor Program** developed by **Microsoft**. Let's use it to create a self-introduction page.

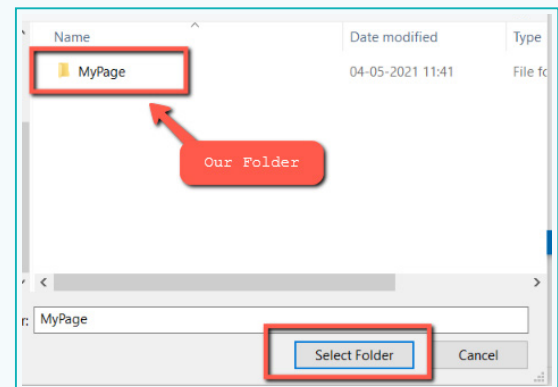
Activity

To start, create a folder called “**MyPage**”, to save HTML and CSS files.

Note:

This folder can be created in any location on the hard drive.

- 1 Open VS Code. Under the **File menu**, click on **Open Folder** and then select the folder we just created.



- 2 Inside the folder, create a file **index.html**. write “**Introduction to HTML and CSS**” inside the title tag. The text you Write within the **<title>** tag is displayed in the title bar of the browser.

```
MyPage > <> index.html > ...
1  <html>
2    <head>
3      <title> Introduction to HTML and CSS </title>
4    </head>
5
6
7  <body>
8
9  </body>
10 </html>
11
```

`<h1>` to `<h6>` are heading tags that display the text inside them as headings. There are 6 heading elements from ‘h1’ to ‘h6’ with descending text sizes.

Hello World

Hello World

Hello World

Hello World

Hello World

Hello World

```
HTML 12unsaved changes X
1 <h1>Hello World</h1>
2 <h2>Hello World</h2>
3 <h3>Hello World</h3>
4 <h4>Hello World</h4>
5 <h5>Hello World</h5>
6 <h6>Hello World</h6>
```

Activity

<p> </p>tag - A paragraph element is used to add a paragraph of text. It has a start and an end tag.

- Inside the body tag, add your name and some basic information about yourself using **heading tags, h2 and h4**. Add a short paragraph about your interests using **<p>** tag.

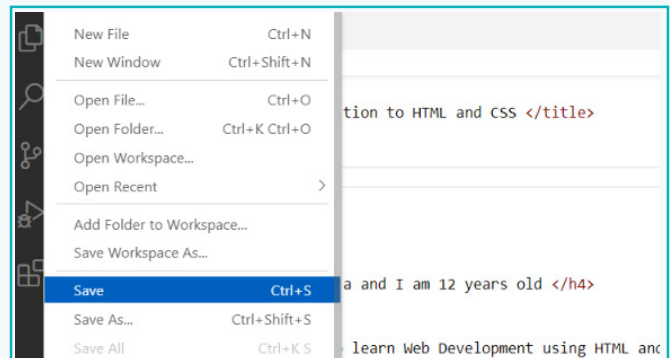
```
<body>
  <h2> Hi, </h2>
  <h4> My name is Anita and I am 12 years old </h4>
  <p>
    I am starting to learn Web Development using HTML and CSS and I
    love designing beautiful web sites
  </p>
</body>
```

Heading tags

Paragraph Tag

Saving an HTML File

Once we are done adding content, we need to save the file. To save a file, use the shortcut Ctrl + S or click on the File menu at the top of VS Code and select Save. Add a name while saving the file. Let's save this with the name **index.html**.



Note:

HTML files are saved using the .html file extension.

- To view the output, go to the **MyPage** folder and double click on **index.html**.

```
<body>
  <h2> Hi, </h2>
  <h4> My name is Anita and I am 12 years old </h4>
  <p>
    I am starting to learn Web Development using HTML and
    CSS and I love designing beautiful web sites
  </p>
</body>
```

Hi,

My name is Anita and I am 12 years old

I am starting to learn Web Development using HTML and CSS and I love designing beautiful web sites

Notice that the output is in black and white, which looks dull. To design the site to look attractive, we will use another language called CSS.

Exercise

 Tick mark ☒ the correct answer

1 Which is the primary html tag used to create a webpage?

a. `<p> ... </p>` ☐

b. `<head> ... </head>` ☐

c. `<html> ... </html>` ☐

d. `<body>... </body>` ☐

2 Which heading tag displays text in the largest size?

a. h1 ☐

b. h2 ☐

c. h6 ☐

d. h4 ☐

 State whether True or False

3 The `<head>` tag is used to display webpage content.

 _____

4 Correct syntax for a paragraph tag is `<p>` content of paragraph `</p>`.

 _____

5 It is compulsory for all elements in HTML to have opening and closing tags.

 _____

4. Stylesheet

Introduction to CSS

With HTML, we create a basic webpage structure and provide the content for it. However, HTML does not have much to offer when it comes to webpage design. For this, we have CSS.

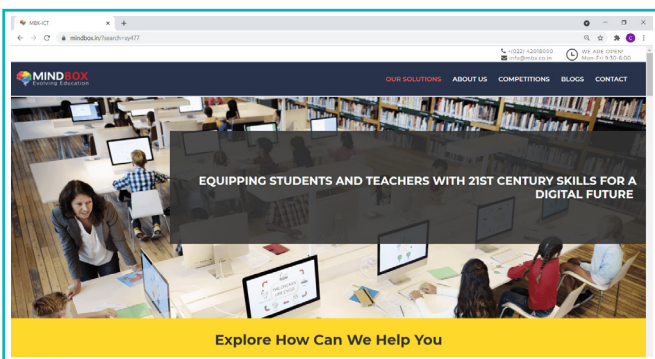
CSS (Cascading Style Sheet) is a language used to design webpages. CSS offers an opportunity to play with the layout of a page, modify colours, fonts, text size, animations, effects, and more.

Here is an example of website designed using CSS:

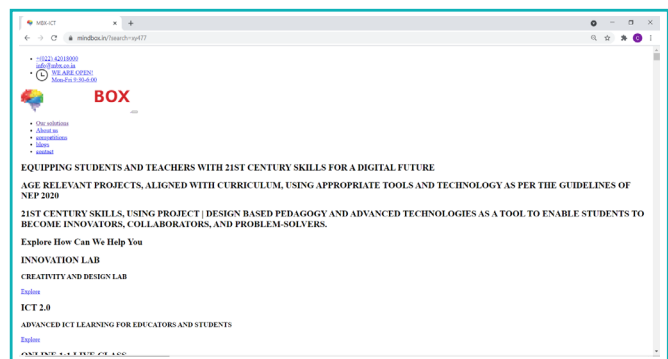


The importance of CSS in web design

Why is CSS important? Look at the two images given below. One is a webpage which uses CSS. The other image is of the same webpage, without CSS.



With CSS



Without CSS

Styling a webpage improves its readability, overall attractiveness and its look and feel. It makes a webpage welcoming and pleasing to browse.

We have understood the purpose and importance of CSS. Now, let's look at how to use CSS to style our webpage.

We can include CSS designs on a web page using 3 simple ways. They are called **Inline**, **Internal** and **External CSS**. In this lesson, we will use the inline method.

Inline CSS

In inline CSS, we add CSS code inside an HTML element in the form of attributes. The CSS code is written within a specific HTML element using the style attribute.

Syntax:

<start-tag style= “property: value”> content </end-tag>

- The attribute “**style**” specifies the CSS code to apply on a webpage.
- **property** refers to style elements like colour, font-family, font-size, etc.
- **value** is used to set the property. For example: the color attribute can have values like red, blue, green etc.

For example:

`<p style = “color: red”> This is an Introduction to CSS! </p>`

In this code, the style attribute specifies that all text written inside the `<p>` tag should be **red** in colour. The declaration specified is referred to as **CSS property**.

This is an Introduction to CSS

Output

To better understand how HTML and CSS work together, let’s work on our webpage and make it look attractive and appealing.

Let’s add some colour to the webpage we have coded in HTML. We will use CSS color properties.

CSS color property

The color property specifies the colour of text, including headings and paragraphs on the web page.

Activity

Apply the color property using the inline CSS method

- 5 Add style attribute in h2 tag and set color property to “blue”.
- 6 Follow punctuation marks “” and : as shown below.

```
<body>  
  <h2 style="color: blue;"> Hi, </h2>  
  <h4> My name is Anita and I am 12 years old </h4>  
  <p>  
    I am starting to learn Web Development using HTML and  
    CSS and I love designing beautiful web sites  
  </p>
```

Hi,

My name is Anita and I am 12 years old

I am starting to learn Web Development using
I love designing beautiful web sites

Activity

- 7 Repeat the same process for the `<h4>` tag and the `<p>` tag, but with different colours.

```
<h2 style="color: blue;"> Hi, </h2>

<h4 style="color: grey;"> My name is Anita and I am 12
years old </h4>

<p style="color: orange;">
  I am starting to learn Web Development using HTML and
  CSS and I love designing beautiful web sites
</p>
```



We have written some text about our interests and we want to change its font style. Let's apply the font "Lucida Sans" to our text.

We need to use a new element called **Division**, which is created using the `<div> .... </div>` tag. A division element can be considered a container, inside which we contain several other HTML elements. It is mainly used to structure layouts or to group related content together.

`<div>`

Content or other HTML elements go here

`</div>`

Create a div tag and enclose headings and paragraphs inside it. Follow the image below:

```
<div>
  <h2 style="color: blue;"> Hi, </h2>
  <h4 style="color: grey;"> My name is Anita and I am 12 years old </h4>
  <p style="color: orange;">
    I am starting to learn Web Development using HTML and CSS and I love
    designing beautiful web sites
  </p>
</div>
```



Changing the font family

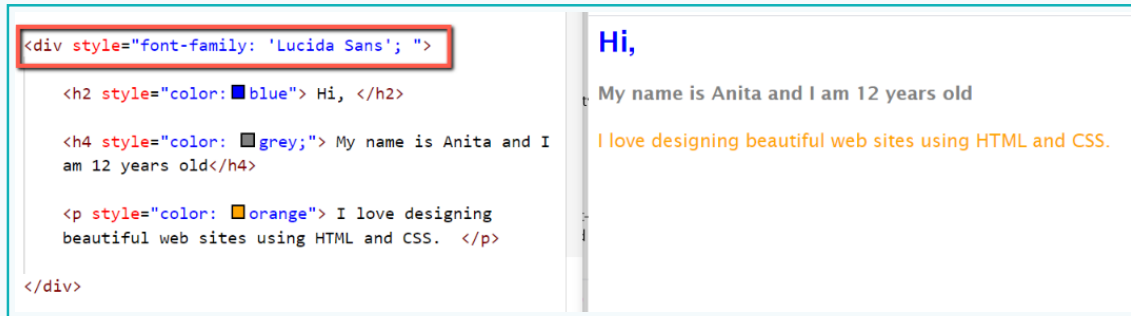
font-family - This property allows us to specify the type of font to use in a webpage.

Applying a font family to content

- 8 Add a **style attribute** in `<div>` tag and set **font-family** to "Lucida Sans".

Activity

- 9 Refresh the browser (using the f5 key) and observe whether the font has changed.



The webpage content is displayed on the left side of the browser window. We will align our content to the centre of the page.

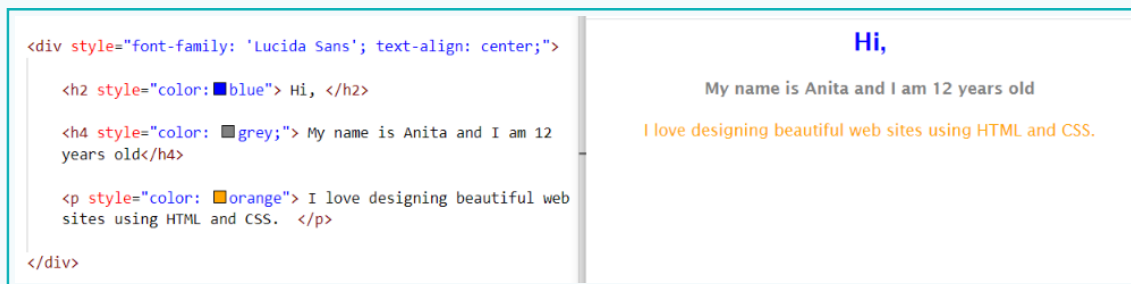
Text Alignment

text-align: A property used to align text inside a div container.

Within the existing CSS property inside the <div> tag, add another property **text-align: center.**

Note:

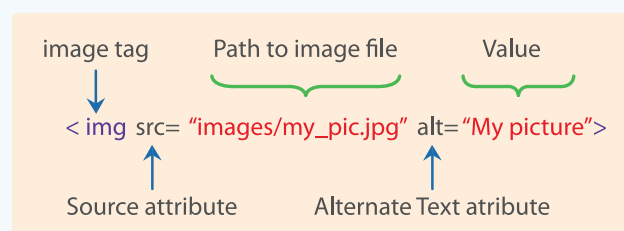
We can add multiple attribute properties to the same element by separating them with a semi-colon (;).



The content is aligned to the centre of the browser now.

Image tag - is a self-closing tag used to display images on a webpage. Source attribute 'src' is used to add the image path. 'alt' attribute is used to add alternate text. Alt text is displayed instead of the image if there is a problem loading.

For example:



Activity

A profile page is incomplete without a picture. Let's add an image to our webpage. We will use the `<img>` tag and specify the file name within the `src` attribute.

- 10 Inside the `<div>` tag, add an `<img>` tag and write the image name in the `src` attribute.

```
<div style="font-family: 'Lucida Sans'; text-align: center;">

  <h2 style="color: blue;"> Hi, </h2>

  <h4 style="color: grey;"> My name is Anita and I am 12
  years old</h4>

  <p style="color: orange;"> I love designing beautiful web
  sites using HTML and CSS. </p>

</div>

</body>
</html>
```

Hi,

My name is Anita and I am 12 years old

I love designing beautiful web sites using HTML and CSS.



The image tag works fine, but the image size is too large. It does not look proportionate to the text. Let's fix that.

Adjust size using CSS

CSS provides two properties, **width** and **height**, to set the dimensions of a container or an image.

- 11 **Width** – Sets the width of a container
- 12 **Height** – Sets the height of a container

Use inline CSS to set the width and height of the image

- 13 Inside the `<img>` tag, apply the style property and set the **width** and **height**.

```
<div style="font-family: 'Lucida Sans'; text-align: center;">

  <h2 style="color: blue;"> Hi, </h2>

  <h4 style="color: grey;"> My name is Anita and I am 12
  years old</h4>

  <p style="color: orange;"> I love designing beautiful web
  sites using HTML and CSS. </p>

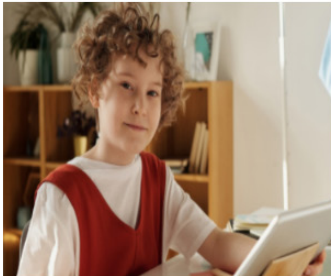
</div>

</body>
</html>
```

Hi,

My name is Anita and I am 12 years old

I love designing beautiful web sites using HTML and CSS.



Activity

Our image now looks proportionate to the text.

Let's add a div container and write a brief description in it.

- 14 Create a div tag and inside, add a <h2> tag and a <p> tag. Use these to add a heading and a small description.

```
<div>
  <h2> About Me </h2>

  <p> I am a student and seek to become a professional Web Developer. Till now, I have created a few websites using HTML and CSS. With every passing day, I am increasing my knowledge on various web technologies so that I can create more complex websites and work as a professional Web Developer.
</p>
</div>
```

About Me

I am a student and seek to become a professional Web Developer. Till now, I have created a few websites using HTML and CSS. With every passing day, I am increasing my knowledge on various web technologies so that I can create more complex websites and work as a professional Web Developer.

Centre align the content using “text-align: center” property. Add a background colour using the “background-color” property.

```
<div style="text-align: center; background-color: lightblue;">
  <h2> About Me </h2>

  <p> I am a student and seek to become a professional Web Developer. Till now, I have created a few websites using HTML and CSS. With every passing day, I am increasing my knowledge on various web technologies so that I can create more complex websites and work as a professional Web Developer.
</p>
</div>
```

About Me

I am a student and seek to become a professional Web Developer. Till now, I have created a few websites using HTML and CSS. With every passing day, I am increasing my knowledge on various web technologies so that I can create more complex websites and work as a professional Web Developer.

- 15 Increase the font size of the paragraph using the “font-size” property

```
<div style="text-align: center; background-color: lightblue;">
  <h2> About Me </h2>

  <p> I am a student and seek to become a professional Web Developer. Till now, I have created a few websites using HTML and CSS. With every passing day, I am increasing my knowledge on various web technologies so that I can create more complex websites and work as a professional Web Developer.
</p>
</div>
```

About Me

I am a student and seek to become a professional Web Developer. Till now, I have created a few websites using HTML and CSS. With every passing day, I am increasing my knowledge on various web technologies so that I can create more complex websites and work as a professional Web Developer.

Activity

FINAL OUTPUT:



In this lesson, we learnt how to create a simple web page from scratch. We added and arranged contents on it using HTML. Then we learnt how CSS properties can be used to apply basic designs elements. We can use CSS to change text and background colour, align contents, and set the width and height of components.

Exercise

■ Fill in the blanks

- 6 The _____ attribute is used to apply inline CSS.
- 7 If multiple CSS properties are specified inside one style tag, they must be separated using a _____ symbol.
- 8 The _____ CSS property is used to set or change the colour of texts on a web page.
- 9 The _____ CSS property is used to set or change the font of a text in a web page.

CSS is a style sheet language which is used to design a webpage. It helps us control the aesthetics of a webpage.



Tech Flash

- **Webpage** : A document created using HTML that can only be viewed in a browser.
- **Website** : A collection of web pages with related content that is identified by a common domain name.
- **HTML** : Short for Hyper Text Markup Language. It is a language used to create the basic structure and layout of a webpage.
- **Tags** : Pre-defined sets of words used to add content in a webpage
- **CSS** : Short for Cascading Style Sheet. It is a language used to design web pages with various in-built properties.